



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO RPS FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

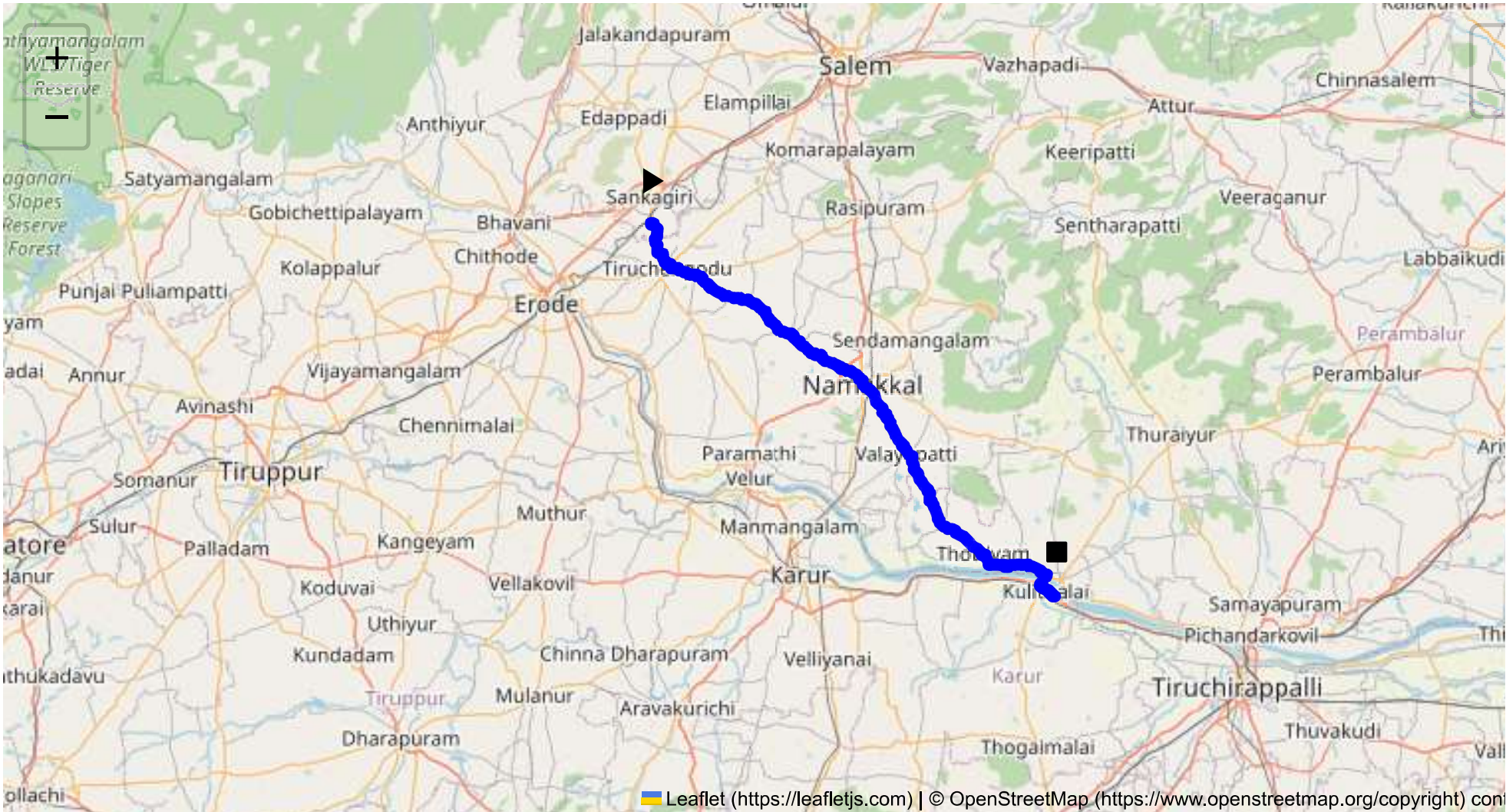
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 93.23 km
Estimated Duration: 2.0 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours
Start: (11.4381, 77.8734)
End: (10.929241, 78.434603)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to Rajendram North spans approximately 93.23 kilometers and typically follows major roadways that interconnect towns and areas between these locations. The journey primarily traverses state highways and passes through semi-urban and rural regions, which include intersections with local roads and potential bottlenecks in small towns.

2. Typical Weather Conditions and Potential Weather-related Hazards: The region experiences a tropical climate with a hot and dry summer, a heavy monsoon season from June to October, and a mild winter. During the monsoon, heavy rains can lead to flooding and waterlogged roads, which might increase the risk of accidents or delays. Drivers should monitor weather forecasts and be prepared for sudden downpours, reduced visibility, and slippery road conditions.

3. Analysis of Traffic Patterns, Highlighting Peak Hours and Congestion-prone Areas: Traffic typically picks up during early morning hours (7 AM - 9 AM) and late afternoon (5 PM - 7 PM), particularly near towns and junctions. Congestion can occur around market areas or small towns like Kumarapalayam and Namakkal, especially when local shops and schools influence traffic flow. Weekends could see heavier traffic due to local activities and events.

4. Assessment of Road Quality and Infrastructure: While major road segments are generally well-maintained, certain stretches might have uneven surfaces or potholes, especially after the rainy season. Roads in rural or less-developed areas may lack thorough signage or adequate lighting, posing a hazard during night-time travel.

5. Suggestions for Alternative Routes for Emergencies: In case of severe weather or road closures, alternative routes may include bypassing smaller towns for highways, though these may add distance or time to the journey. Route via Namakkal and Trichy road offers an alternate path should primary roads become non-navigable.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Transport of hazardous materials requires adherence to state and national regulations, including proper vehicle signage, driver certifications, permissible times for transport (avoiding peak traffic in city limits), and certain routes designated specifically for such materials to limit exposure near populated areas.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials: The region has a moderate history of vehicular accidents, predominantly minor collisions near city centers and junctions. There are limited reported incidents specifically involving hazardous materials, but general traffic accidents suggest attention to road intersections and speed regulations.

8. Environmental Considerations and Sensitive Areas: The route crosses agricultural lands and near several water bodies. Drivers should be cautious of wildlife areas or natural reserves near the route that might lead to road crossings by animals, more so during early morning or evening hours.

9. Analysis of Communication Coverage, Noting Potential Dead Zones: Majority of the route is covered by mobile networks except for isolated stretches in rural or low-signal areas. Dead zones are mostly outside city or town centers, suggesting preparation for limited communication in these areas.

10. Estimated Emergency Response Times for Different Route Segments: Emergency response times in urban areas can vary from 15 to 30 minutes depending on traffic, while in rural segments, it might extend to 45 minutes or more. Availability of roadside assistance may be lower in rural areas.

11. Overall Summary of Risk Assessment: The route from Sangagiri to Rajendram North generally poses moderate travel risk. Potential hazards include weather-related disruptions, traffic congestion, varying road conditions, and patchy communication coverage. Critical risk management includes monitoring weather updates, careful planning of travel times to avoid peak congestion, ensuring vehicle

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.94342, 78.41370	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44019, 77.87547	15 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	Medium	11.38342, 77.89975	30 KM/Hr
7	Turn	Medium	11.38295, 77.90001	30 KM/Hr
8	Turn	High	11.38186, 77.90258	15 KM/Hr
9	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
10	Turn	Medium	11.28043, 78.07413	30 KM/Hr
11	Turn	Medium	11.22407, 78.16640	30 KM/Hr
12	Turn	Medium	11.21962, 78.16777	30 KM/Hr
13	Turn	Medium	10.98680, 78.33772	30 KM/Hr
14	Turn	Medium	10.97261, 78.34387	30 KM/Hr
15	Turn	Medium	10.97202, 78.36602	30 KM/Hr
16	Turn	High	10.96016, 78.42212	15 KM/Hr
17	Turn	Medium	10.96069, 78.42248	30 KM/Hr
18	Turn	Medium	10.96084, 78.42307	30 KM/Hr
19	Blind Spot	Blind Spot	10.96050, 78.42371	10 KM/Hr
20	Turn	Medium	10.93907, 78.42275	30 KM/Hr
21	Turn	Medium	10.93915, 78.42388	30 KM/Hr
22	Turn	Medium	10.93449, 78.42786	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
18	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
19	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
20	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
21	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
22	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
23	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
24	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
25	clinic	Dhanvanthiri Physiotheraphy	11.2245256, 78.1638528	30 km/h	Medium
26	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
27	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
28	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium
30	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
31	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium
32	hospital	Aravinth Hospital	11.2233336, 78.1672805	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
33	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
34	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium
35	hospital	Sri Nidhi Hospital, Namakkal	11.2255711, 78.1687291	30 km/h	Medium
36	clinic	Athithya Nursing Home	11.2237726, 78.1686447	30 km/h	Medium
37	hospital	Kathiravan Hospital	11.226651, 78.167351	30 km/h	Medium
38	hospital	Sri Ramana Hospital	11.223503, 78.1685433	30 km/h	Medium
39	hospital	Vanitha Child Care & Skin Hospital	11.2243378, 78.16811	30 km/h	Medium
40	hospital	Sri Dhanvanthries Hospital	11.2241181, 78.1678614	30 km/h	Medium
41	hospital	Nirmala Hospital	11.2179361, 78.1665001	30 km/h	Medium
42	hospital	Surya Hospital, Namakkal	11.2178506, 78.1660375	30 km/h	Medium
43	hospital	Shanthi Hospitals, Namakkal	11.2190661, 78.1685735	30 km/h	Medium
44	hospital	Ganga Hospital	11.21742, 78.170272	30 km/h	Medium
45	hospital	Government Hospital	11.2168924, 78.169068	30 km/h	Medium
47	police	Police Station	11.2185353, 78.1695347	30 km/h	Medium
48	police	Women Police Station	11.2183133, 78.1692677	30 km/h	Medium
49	hospital	Manoharan Hospital	11.218016, 78.171629	30 km/h	Medium
50	hospital	Saravana Hospital	11.216652, 78.171298	30 km/h	Medium
52	hospital	Latha Eye Hospital	11.2185689, 78.1719035	30 km/h	Medium
53	hospital	Vasan Eye Care Hospital, Namakkal	11.216301, 78.1717694	30 km/h	Medium
54	hospital	Sri Ragavendra Hospital	11.2151734, 78.1703431	30 km/h	Medium
55	hospital	Aishwarya Hospital	11.2164042, 78.1715972	30 km/h	Medium
56	hospital	Thangam Hospital,Namakkal	11.21472, 78.171857	30 km/h	Medium
57	hospital	Poornima Hospital	11.215533, 78.16911	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
58	hospital	Nagarajan Hospital	11.2171673, 78.1718618	30 km/h	Medium
59	clinic	Ramesh ENT & Skin Care	11.217133, 78.171071	30 km/h	Medium
60	hospital	SMK Hospital	11.2172759, 78.1706866	30 km/h	Medium
61	clinic	ENT Clinic	11.215786, 78.170545	30 km/h	Medium
62	clinic	Hari Chest Clinic	11.2162053, 78.1710093	30 km/h	Medium
63	hospital	Dr. Kandasamy Hospital	11.216051, 78.172052	30 km/h	Medium
64	clinic	Primecare Diabetic Clinic	11.213992, 78.1737091	30 km/h	Medium
65	clinic	Dr. Shanmugam ENT Clinic	11.218762, 78.171206	30 km/h	Medium
66	clinic	Children's Clinic	11.217004, 78.171504	30 km/h	Medium
67	clinic	Krishiv's Dental Care & Orthodontic Centre	11.2184886, 78.1713346	30 km/h	Medium
71	hospital	Dr. A. Nagarajan Children's Hospital	11.212765, 78.1747068	30 km/h	Medium
72	hospital	M.M. Hospital	11.2103089, 78.1776545	30 km/h	Medium
73	clinic	Ram Clinic	11.2090922, 78.1790782	30 km/h	Medium
74	hospital	Imayam Speciality Hospital	11.2078899, 78.1797861	30 km/h	Medium
75	clinic	Prasith Clinic	11.208902, 78.180957	30 km/h	Medium
76	clinic	Aburvaa Multi Special Herbal Clinic	11.207568, 78.183483	30 km/h	Medium
77	hospital	Sugam Children's Hospital	11.2060083, 78.1854033	30 km/h	Medium
79	hospital	Government Hospital, Valayapatty	11.1212591, 78.2315962	30 km/h	Medium
82	hospital	Saravana Hospitals	10.9944313, 78.3273838	30 km/h	Medium
83	hospital	Dr. Siva Hospital	10.9877652, 78.3359228	30 km/h	Medium
84	hospital	Dr. Alva Hospital	10.987225, 78.336154	30 km/h	Medium
85	hospital	Ravikumar Hospital	10.988317, 78.336248	30 km/h	Medium
86	police	Thottiyam Police station	10.9874415, 78.3363001	30 km/h	Medium
87	hospital	Government Hospital, Thottiyam	10.98388, 78.338648	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
88	hospital	Government Hospital	10.9428055, 78.4146362	30 km/h	Medium
89	hospital	T M R Hospital	10.9429825, 78.4158242	30 km/h	Medium
90	clinic	Raka Nursing Home	10.9386263, 78.4244032	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
16	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium
17	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium
29	school	Government Girls School	11.2238954, 78.1662583	30 km/h	Medium